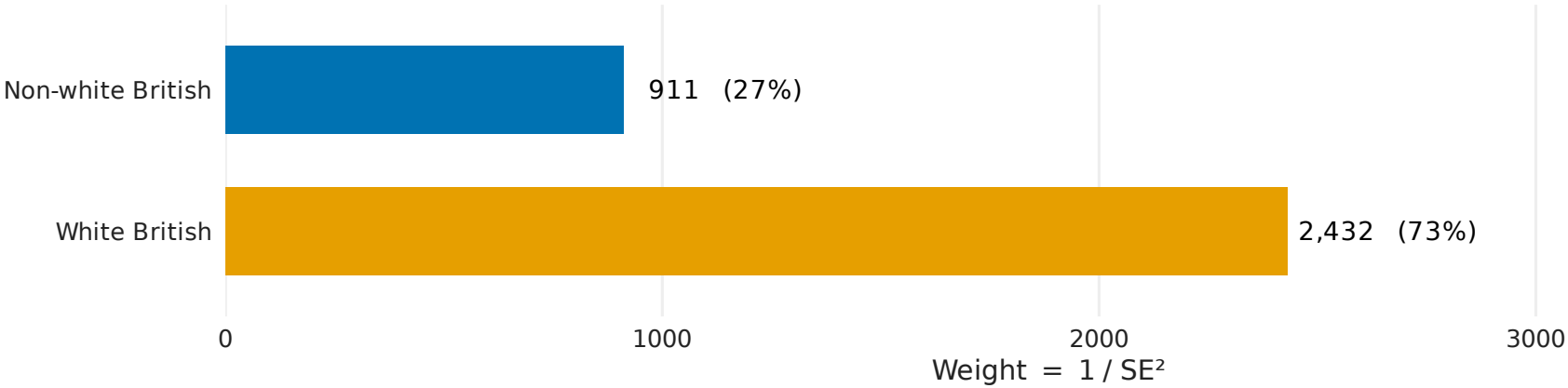
3. Inverse-variance weighting is a balance, not a vote

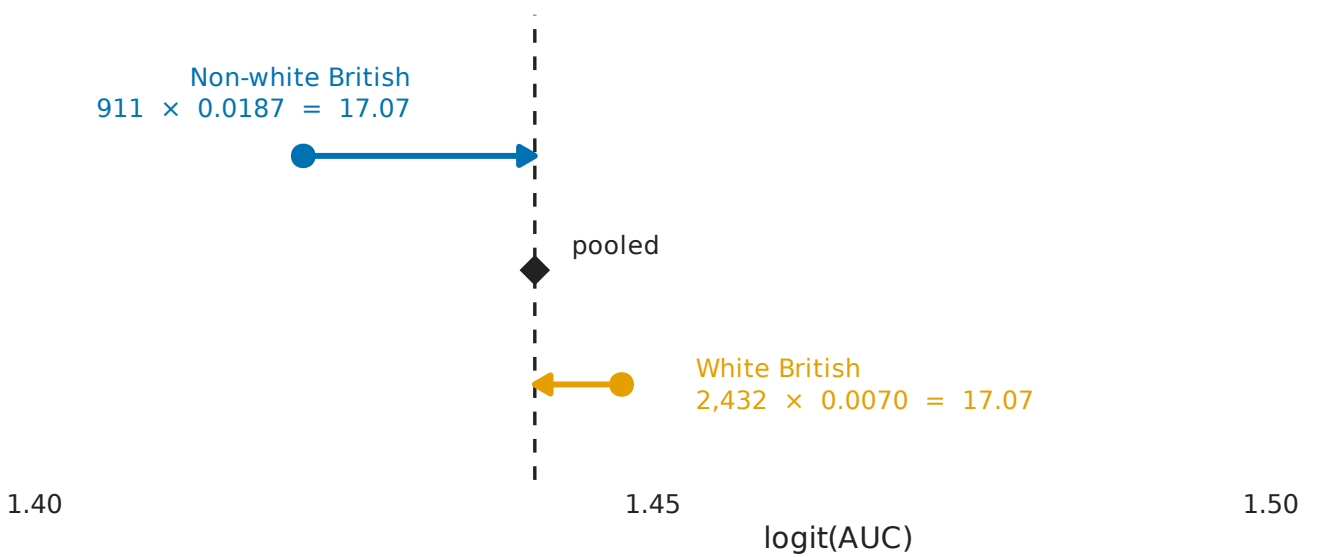
Weight = 1 / SE². The white British evaluation is larger and has the tighter interval, so it holds about 73% of the weight. The common value moves toward that study until weight times distance is the same on both sides.

Weight of each study



The pooled logit is where the pulls cancel

Both products equal 17.07. The heavier study sits closer to the dashed line.



Pooled logit = (w1 × eta1 + w2 × eta2) / (w1 + w2) = 1.440. SE of that mean = 1 / sqrt(sum of weights) = 0.0173. AUC = plogis(1.440) = 0.809.